ECE 3317

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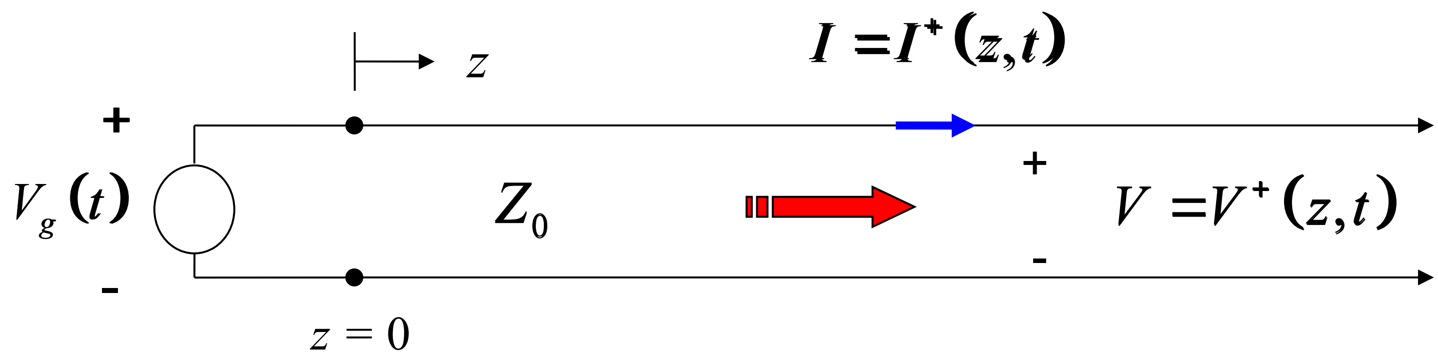
Notes 7
Transmission Lines
(Pulse Propagation and Reflection)

Adapted from notes by Prof. Stuart A. Long

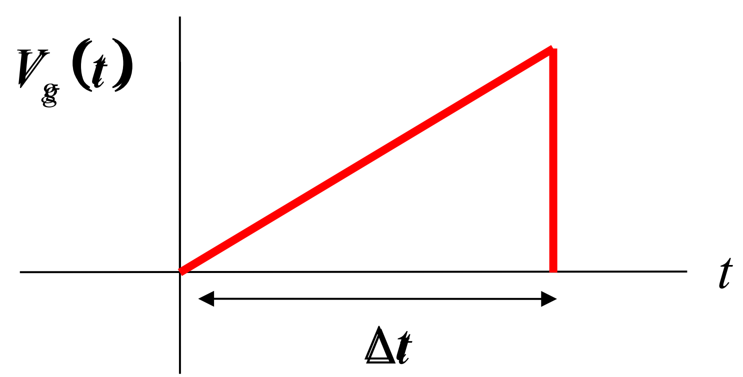
Pulse on Transmission Line

A voltage signal is applied at the input of the semi-infinite transmission line.

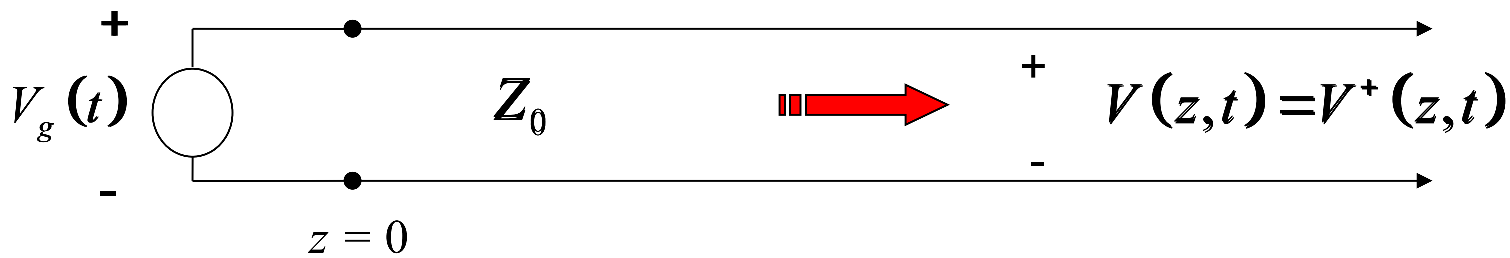


Example:

Sawtooth wave



Pulse on Transmission Line (cont.)



$$V(z,t) = f(z - c_d t)$$

Goal: determine the function f

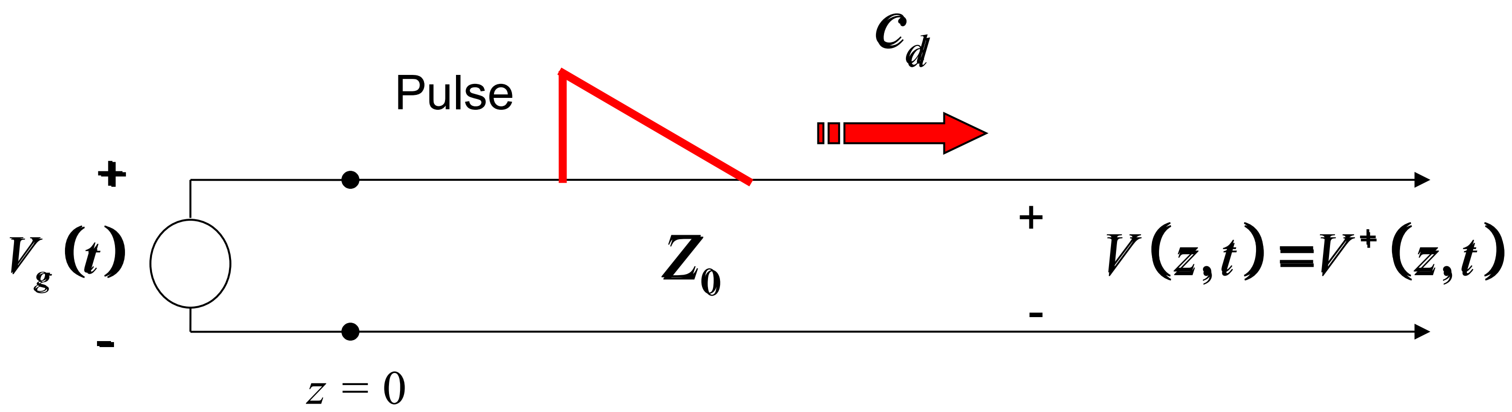
At $z = 0$: $V(0,t) = f(-c_d t) = V_g(t)$

$$\Rightarrow \quad f(\zeta) = V_g(-\zeta / c_d) \qquad (\zeta = -c_d t)$$

Hence

$$V(z,t) = f(z - c_d t) = V_g(-[z - c_d t]/c_d) = V_g(t - z/c_d)$$

Pulse on Transmission Line (cont.)



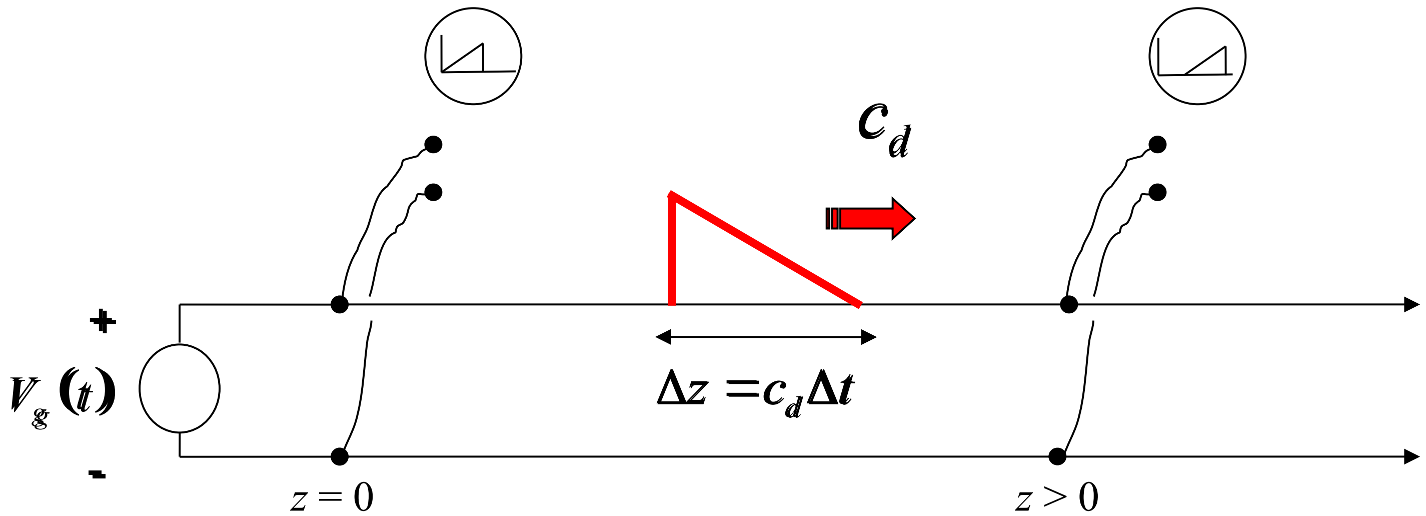
$$V(z,t) = V_g(t - z/c_d)$$

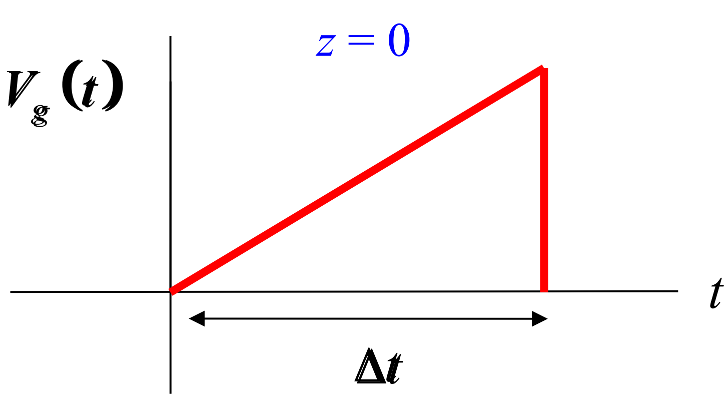
At any position z, the pulse that is measured is the same as the input pulse, except that it is delayed by a time $t_d = z/c_d$.

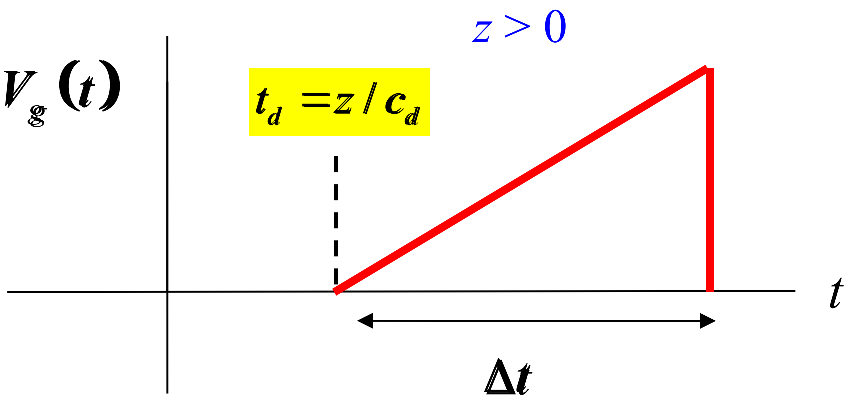
Note that the shape of the pulse as a function of z is a scaled mirror image of the pulse shape as a function of t.

Pulse on Transmission Line (cont.)

Note the delay in the trace on this oscilloscope.



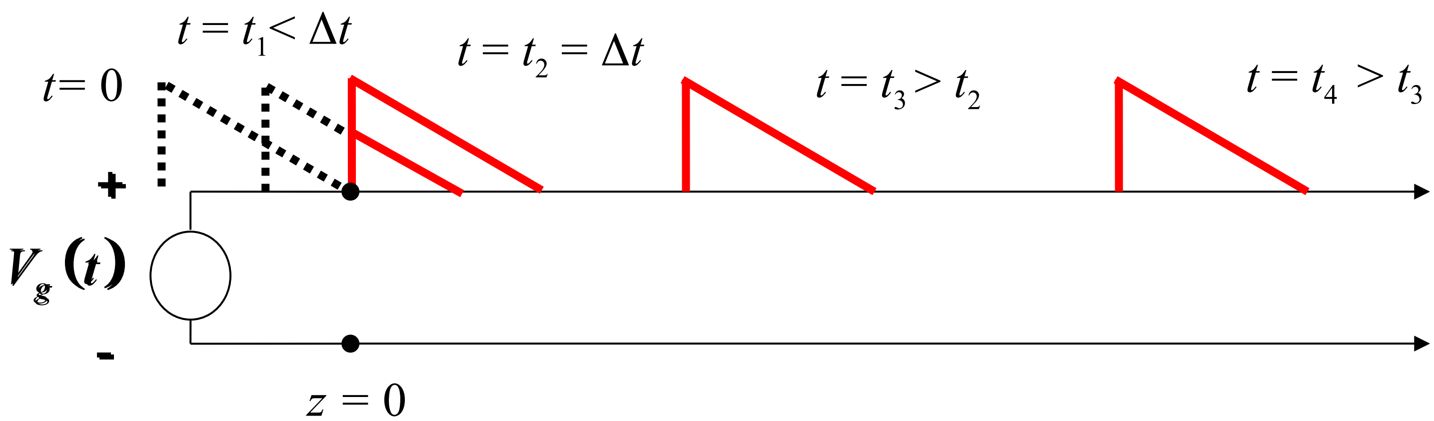


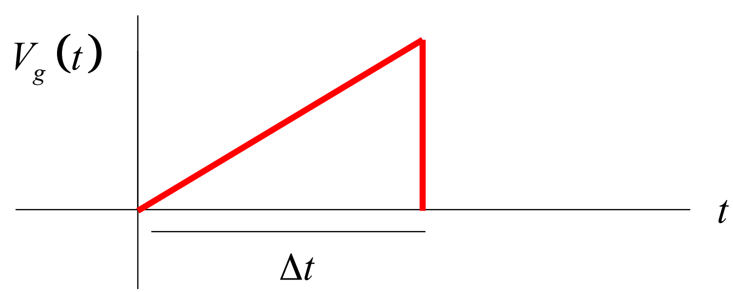


Pulse on Transmission Line (cont.)

The pulse is shown emerging from the source end of the line.

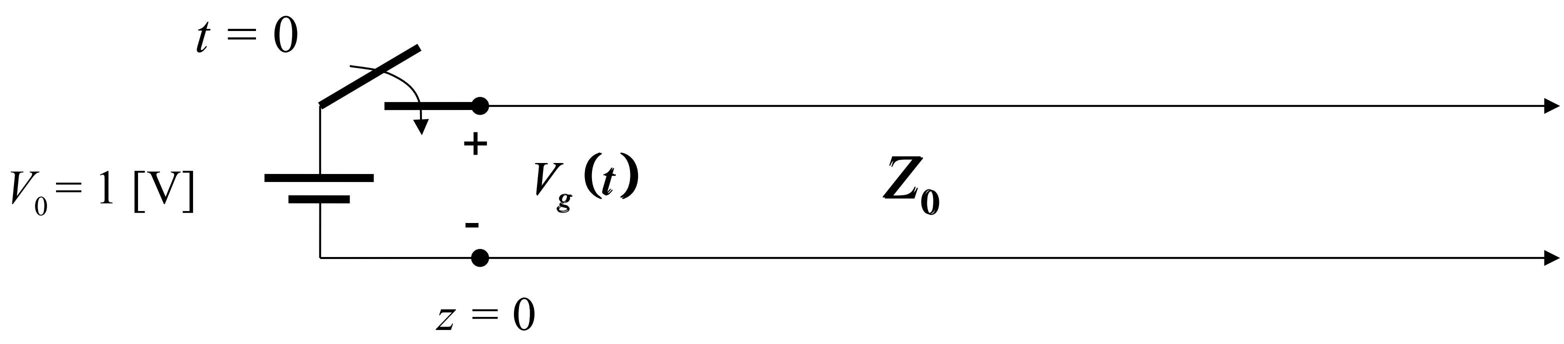
A series of “snapshots” is shown.

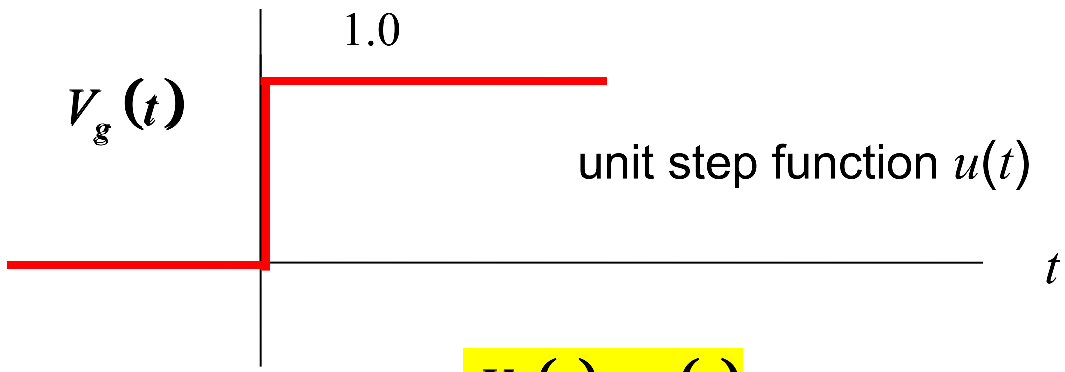




Step Function Source

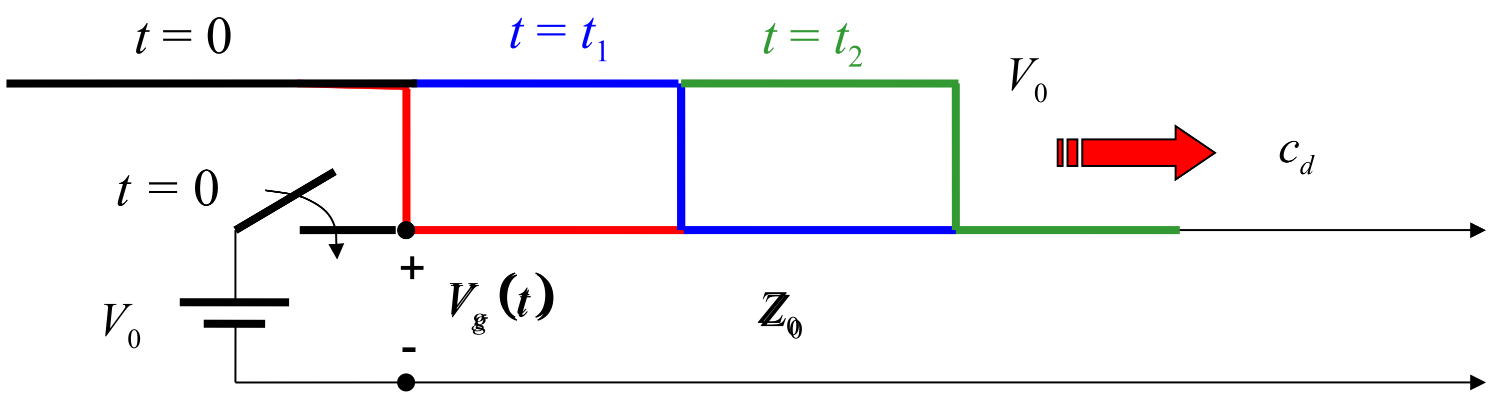
Another example (battery and switch)





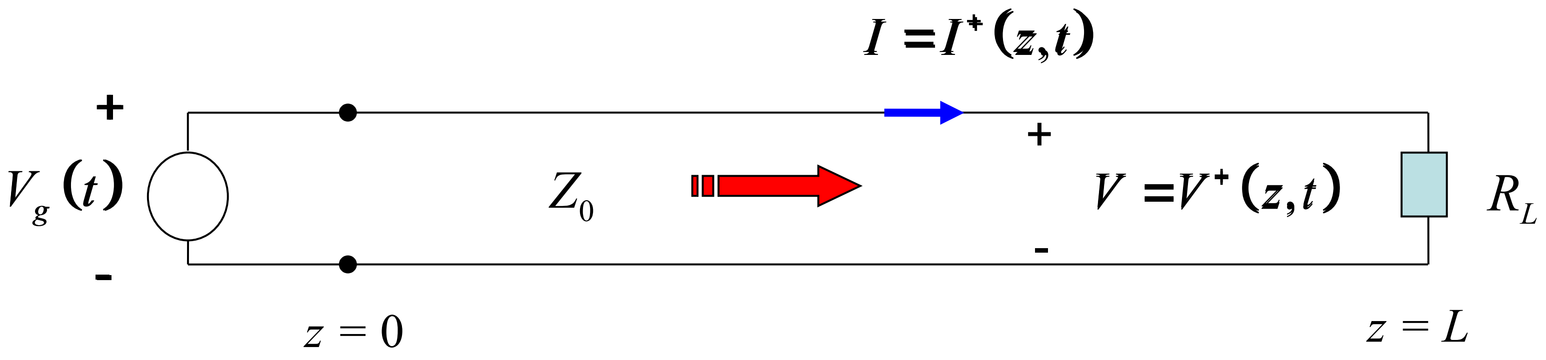
$$V_g(t) = u(t)$$

Step Function Source (cont.)



Matched Load

$$R_L = Z_0$$



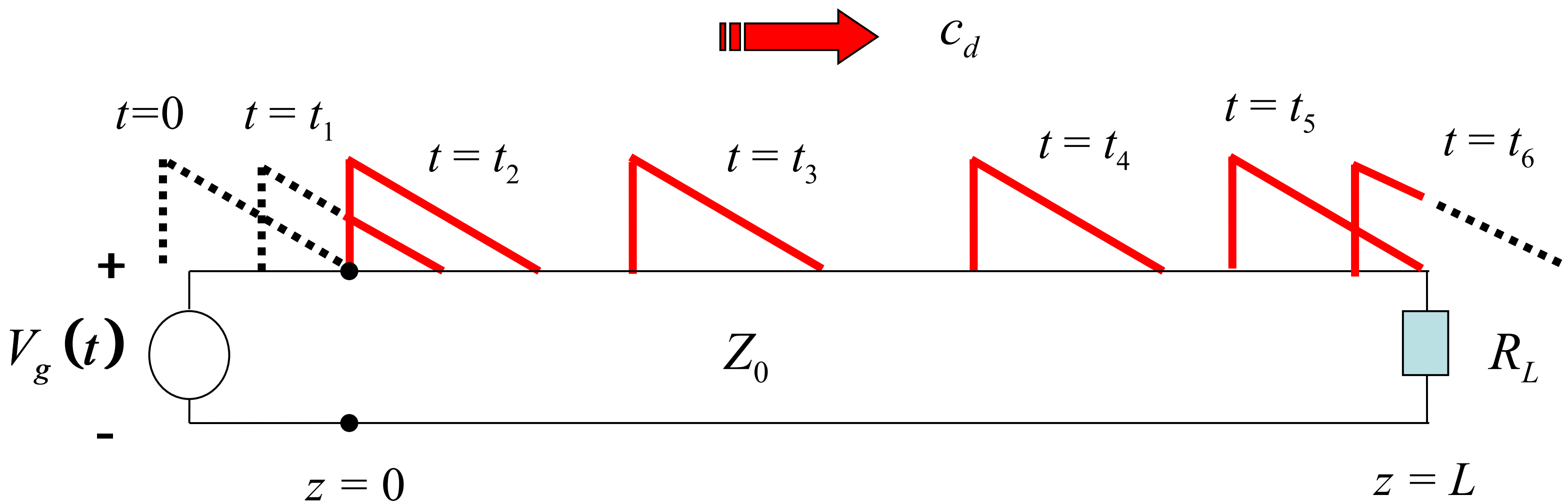
On line: $$\frac{V^+(z,t)}{I^+(z,t)} = Z_0$$ (forward-traveling wave)

At load: $$\frac{V(L,t)}{I(L,t)} = R_L$$

Hence, at $z = L$, the two relations are the same since $R_L = Z_0$.
When the waveform hits the load end, it "sees" a continuation of the line.
There is no reflection.

Absorption by Load

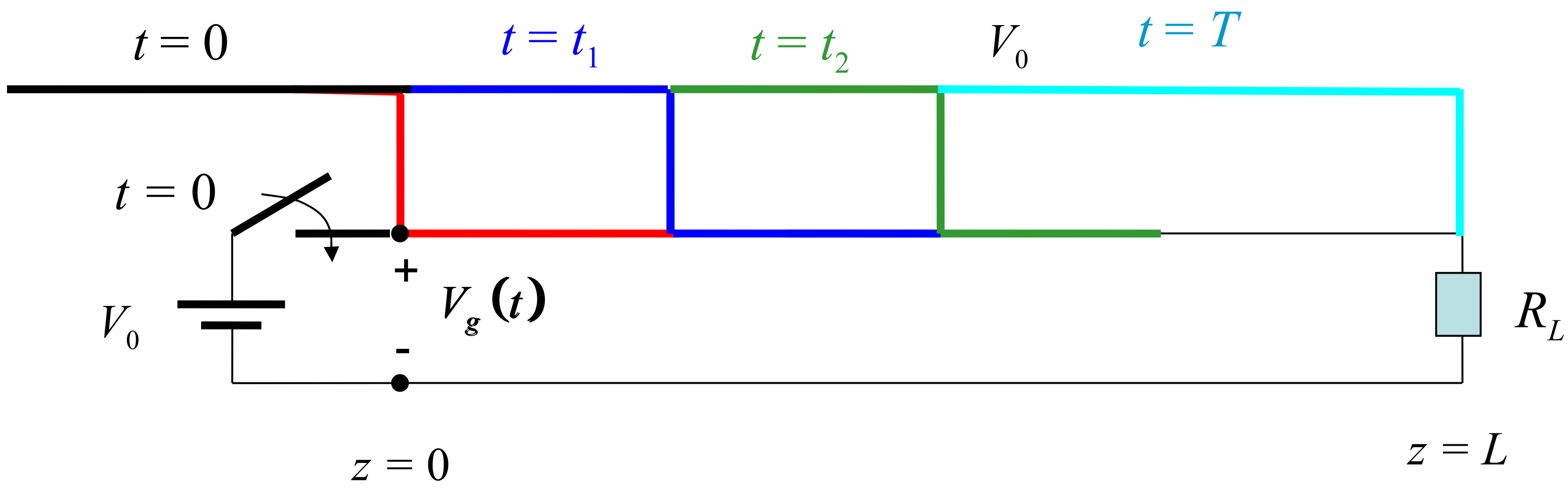
This shows the sawtooth waveform propagating on a matched line.



The pulse is shown emerging from the source end of the line, traveling down the line, and then being absorbed by the matched load.

Absorption by Load (cont.)

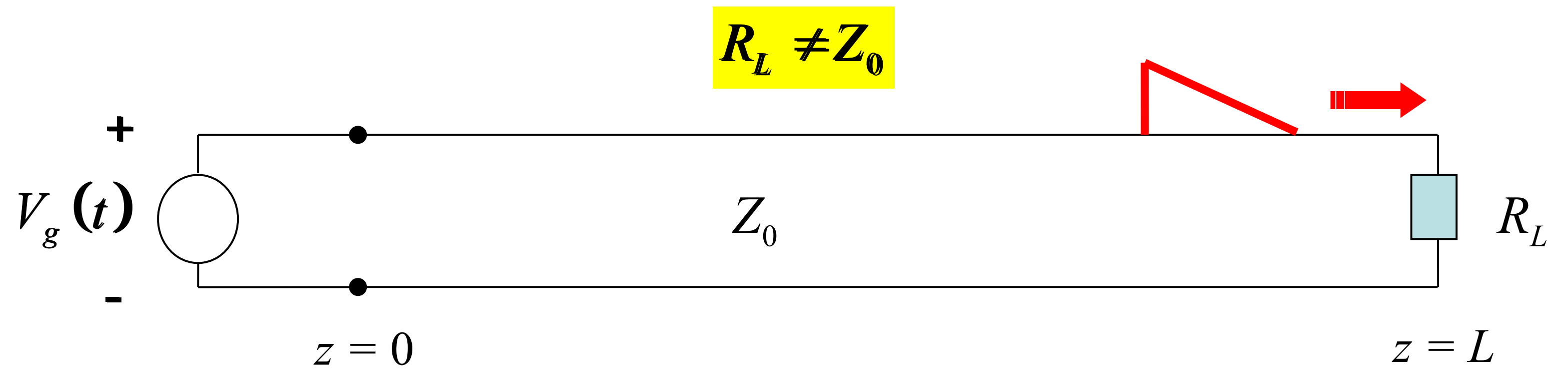
Step function on matched line



Time to reach the load end: $\boxed{T = \dfrac{L}{c_d}}$

For $t > T$ we have reached steady state: $V(z,t) = V_0$ everywhere on the line.

Arbitrary Load



On line:

$$V(z,t) = V^+(t - z/c_d) + V^-(t + z/c_d)$$

$$I(z,t) = \frac{1}{Z_0}\left[V^+(t - z/c_d) - V^-(t + z/c_d)\right]$$

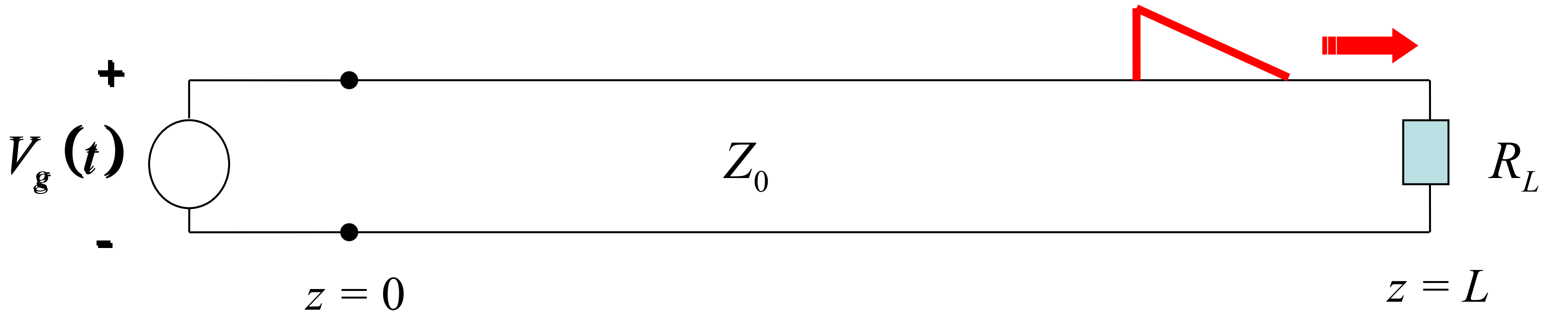
where $V^+(t - z/c_d) = V_g(t - z/c_d)$ (known function)

Goal: solve for $V^-(z,t)$

At load: $\dfrac{V(L,t)}{I(L,t)} = R_L$

Arbitrary Load (cont.)

$R_L \neq Z_0$



At load: $$\frac{V(L,t)}{I(L,t)} = R_L$$

Hence $$\frac{V^+(t - L/c_d) + V^-(t + L/c_d)}{\frac{1}{Z_0}\left[V^+(t - L/c_d) - V^-(t + L/c_d)\right]} = R_L$$

Arbitrary Load (cont.)

$$V^{+}(t - L/c_d) + V^{-}(t + L/c_d) = R_L\left(\frac{1}{Z_0}\left[V^{+}(t - L/c_d) - V^{-}(t + L/c_d)\right]\right)$$

$$\Downarrow$$

$$V^{-}(t + L/c_d)\left[1 + \frac{R_L}{Z_0}\right] = V^{+}(t - L/c_d)\left[-1 + \frac{R_L}{Z_0}\right]$$

$$\Downarrow$$

$$V^{-}(t + L/c_d) = V^{+}(t - L/c_d)\left(\frac{-1 + \dfrac{R_L}{Z_0}}{1 + \dfrac{R_L}{Z_0}}\right)$$

$$\Downarrow$$

$$V^{-}(t + L/c_d) = V^{+}(t - L/c_d)\left(\frac{R_L - Z_0}{R_L + Z_0}\right)$$

Arbitrary Load (cont.)

$$V^{-}\left(t+L/c_d\right)=V^{+}\left(t-L/c_d\right)\left(\frac{R_L-Z_0}{R_L+Z_0}\right)$$

Define

$$V_L^{+}\left(t\right)=V^{+}\left(t-L/c_d\right)$$ voltage of forward-traveling wave at the load

$$V_L^{-}\left(t\right)=V^{-}\left(t+L/c_d\right)$$ voltage of backward-traveling wave at the load

Then $$V_L^{-}\left(t\right)=V_L^{+}\left(t\right)\left(\frac{R_L-Z_0}{R_L+Z_0}\right)$$

Define $$\boxed{\Gamma_L=\left(\frac{R_L-Z_0}{R_L+Z_0}\right)}$$ load reflection coefficient

We then have $$\boxed{V_L^{-}\left(t\right)=\Gamma_L V_L^{+}\left(t\right)}$$

Arbitrary Load (cont.)

Summary for load reflection

$$V_L^-(t) = \Gamma_L V_L^+(t)$$

$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

load reflection coefficient

Note: There is no reflection for a matched system ($R_L = Z_0$).

Note:

$$-1 \le \Gamma_L \le +1$$

Arbitrary Load (cont.)

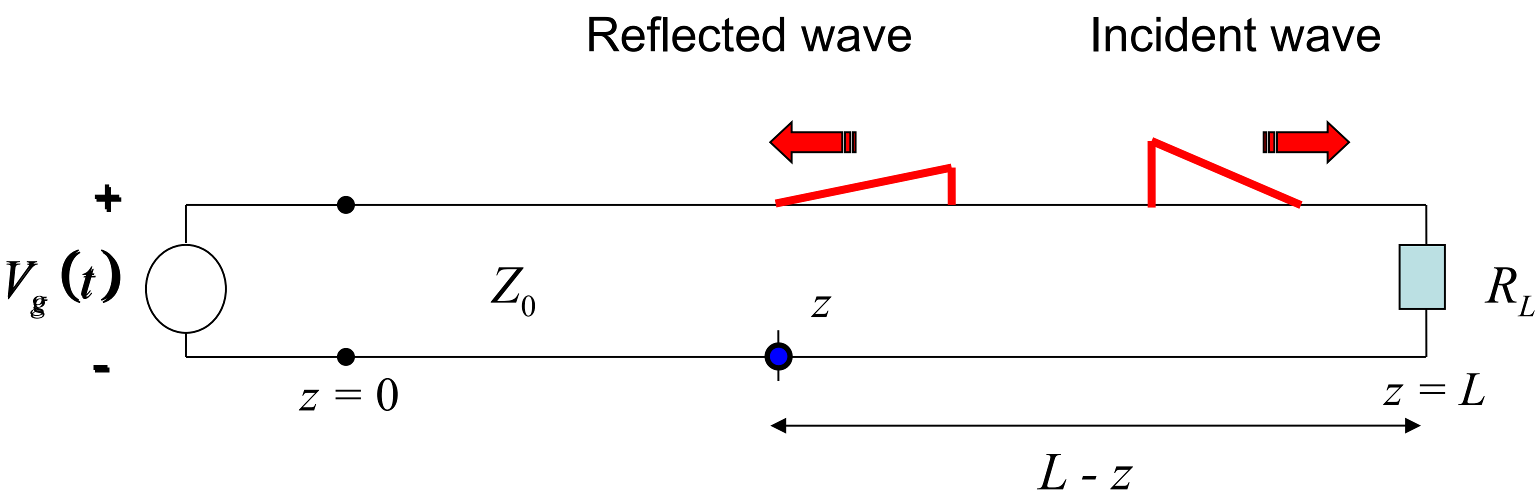
We now proceed to obtain $V^{-}(t + z / c_d)$ (for arbitrary z)

At the load: $V^{-}(t + L / c_d) = V_L^{-}(t)$

The reflected wave is delayed from the load due to the distance $L - z$.

Hence $$V^{-}(t + z / c_d) = V_L^{-}(t - (L - z) / c_d)$$

Reflection Picture



$$V^-(t + z/c_d) = V_L^-\left(t - (L - z)/c_d\right)$$
$$= \Gamma_L V_L^+\left(t - (L - z)/c_d\right)$$

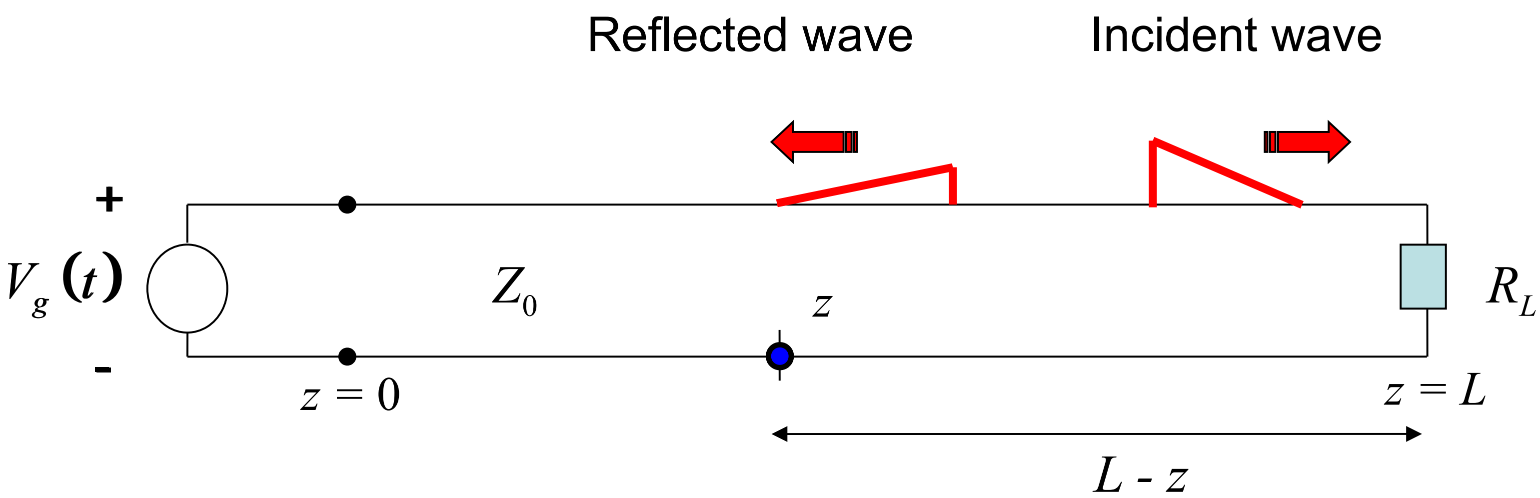
Substitute into argument

Also, we have $V_L^+(t) = V_g(t - L/c_d)$

Hence $V^-(t + z/c_d) = \Gamma_L V_g\left(\left[t - (L - z)/c_d\right] - L/c_d\right)$

Notes:
- The reflected wave is the mirror image of the incident wave, reduced in amplitude.
- The reflected wave is delayed from the load after traveling a distance L-z.

Reflection Picture



Time it takes to go from load to observation point z

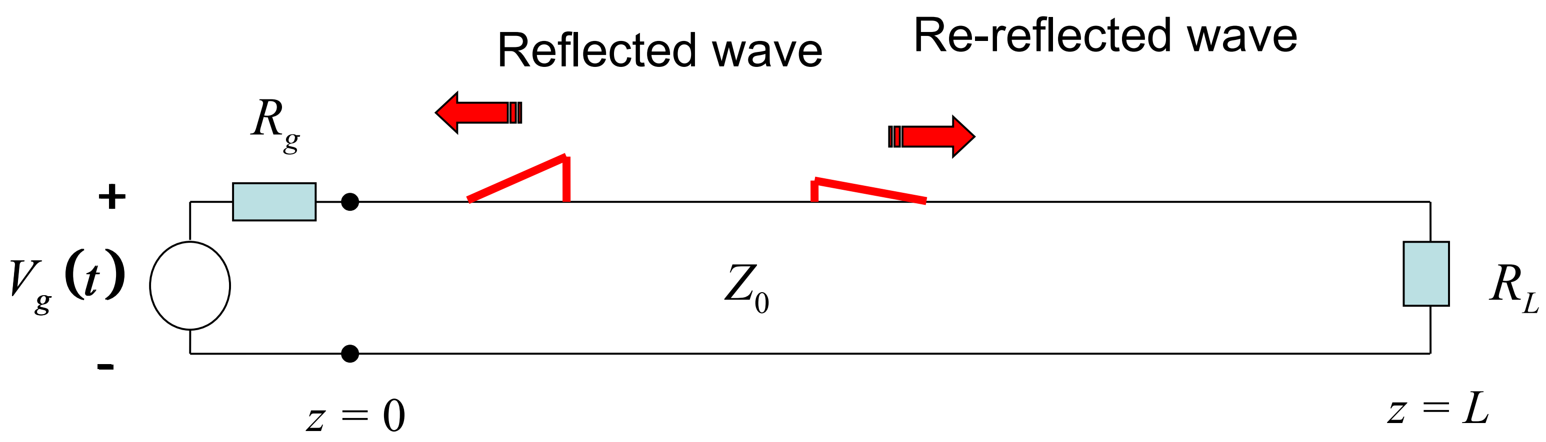
Time it takes to reach load

$$V^-(t + z/c_d) = \Gamma_L V_g\left(\left[t - (L - z)/c_d\right] - L/c_d\right)$$

Reflections at Source End

After the reflected wave hits the source end,
there may be another reflection.

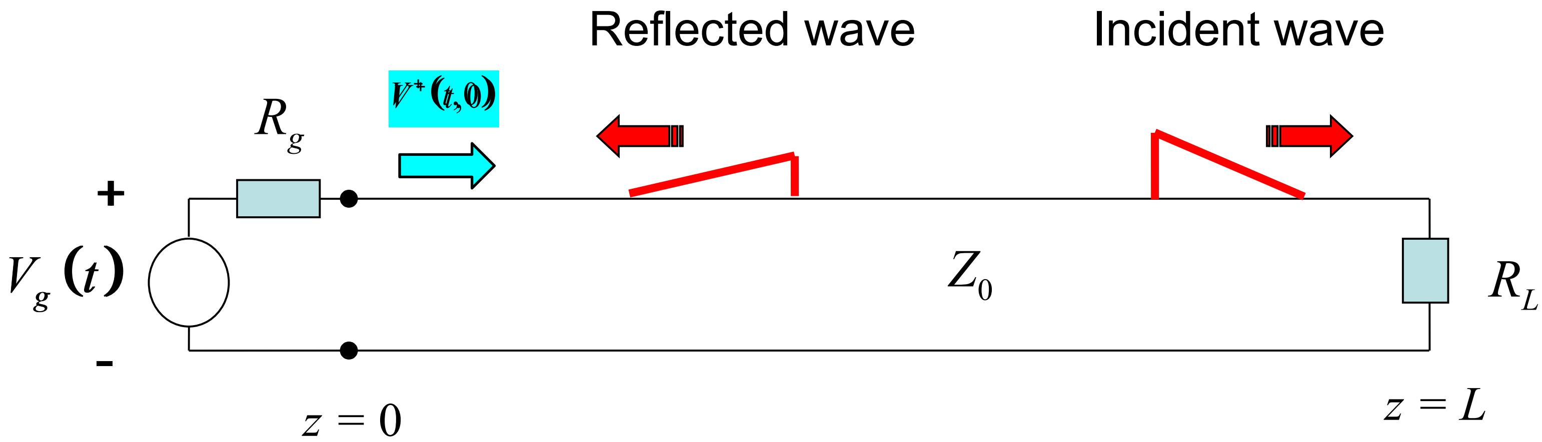
Here we allow the source to have a Thévenin resistance R_g.



$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right)$$

Note: In calculating the reflection from the source, we *turn off* the source voltage.

Complete Wave Picture



Incident: $V(z,t) = AV_g(t - z/c_d)$

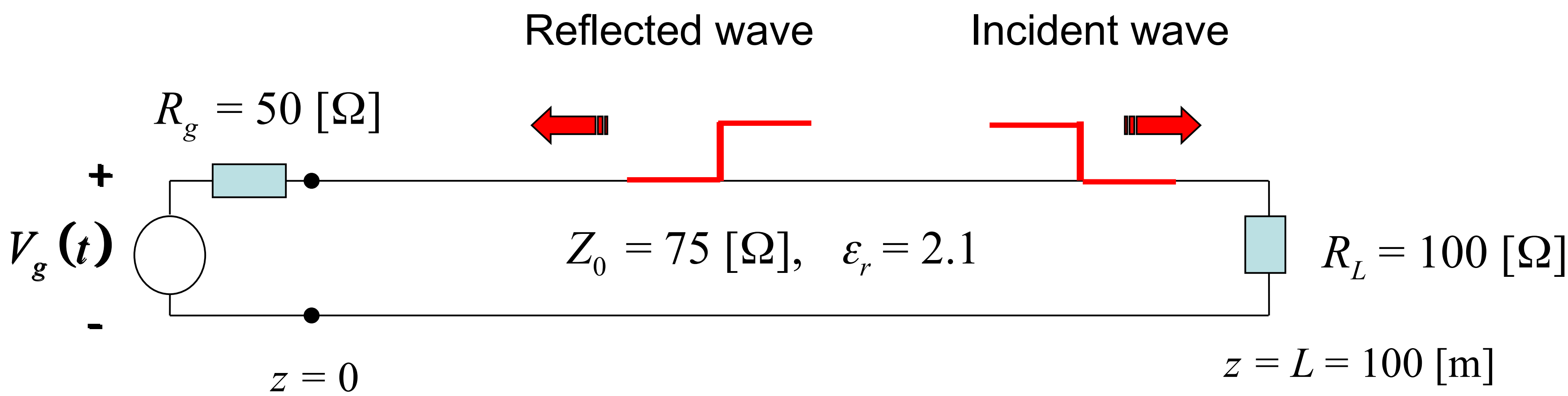
Reflected: $V(z,t) = \Gamma_L AV_g(t - L/c_d - (L - z)/c_d)$

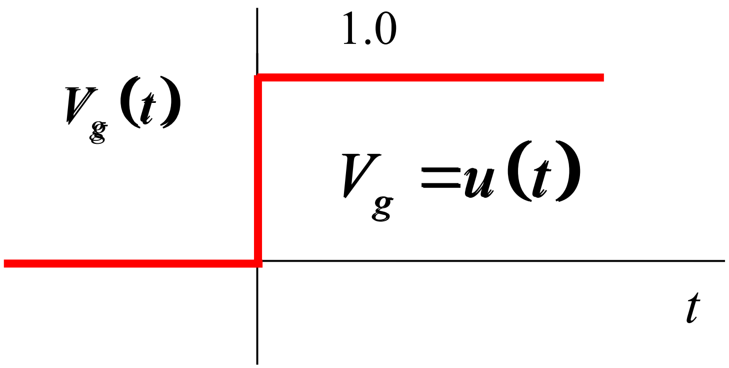
Re-reflected: $V(z,t) = \Gamma_g \Gamma_L AV_g(t - 2L/c_d - z/c_d)$

Re-re-reflected: $V(z,t) = \Gamma_g \Gamma_L^2 AV_g(t - 3L/c_d - (L - z)/c_d)$

We'll stop here …but we could do more reflections.

Example



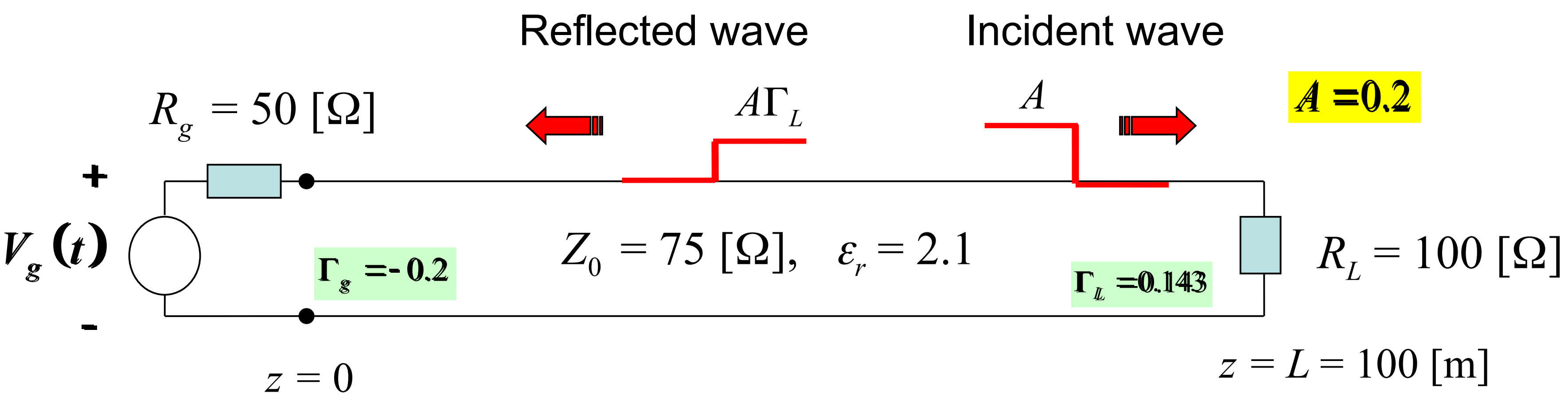


$$c_d = c/\sqrt{2.1} = 2.069 \times 10^8\ [\mathrm{m/s}]$$

$$\Gamma_L = \frac{100 - 75}{100 + 75} = 0.1429$$

$$\Gamma_g = \frac{50 - 75}{75 + 50} = -0.2000$$

Example (cont.)



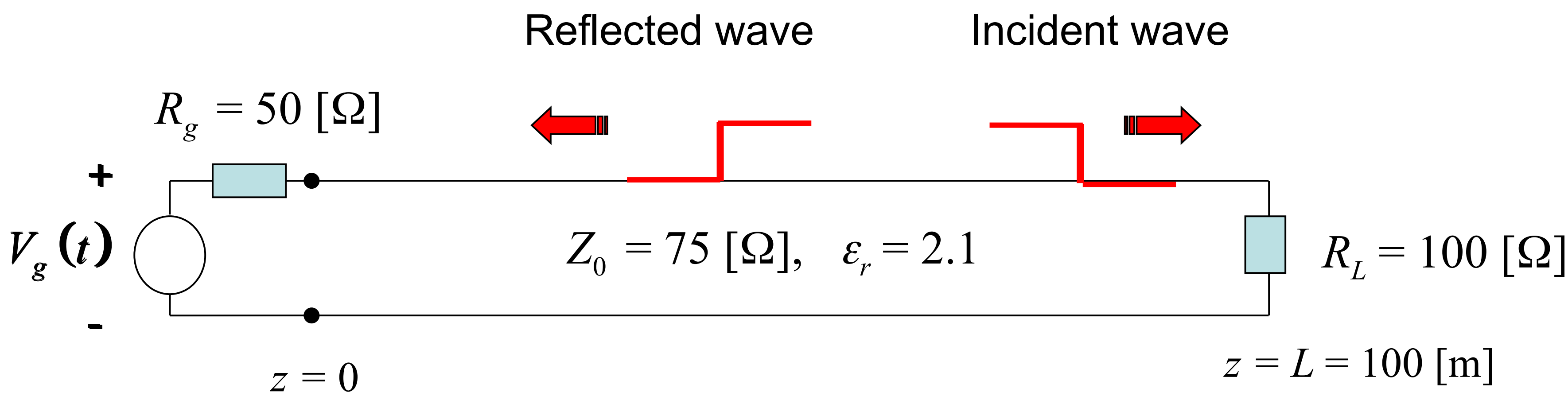
Incident: $V(z,t) = (0.2)\,u(t - z/c_d)$

Reflected: $V(z,t) = (0.143)(0.2)\,u(t - L/c_d - (L - z)/c_d)$

Re-reflected: $V(z,t) = (-0.2)(0.143)(0.2)\,u(t - 2L/c_d - z/c_d)$

Re-re-reflected: $V(z,t) = (0.143)(-0.2)(0.143)(0.2)\,u(t - 3L/c_d - (L - z)/c_d)$

Example (cont.)



Total voltage:

$$
\begin{aligned}
V(z,t) = &(0.2)u(t - z/c_d) \\
&+ (0.0286)u(t + z/c_d - 2L/c_d) \\
&+ (-0.00572)u(t - z/c_d - 2L/c_d) \\
&+ (-0.000817)u(t + z/c_d - 4L/c_d) \\
&+ \ldots
\end{aligned}
$$

$L = 100\ [\text{m}]$

$c_d = c/\sqrt{2.1} = 2.069 \times 10^8\ [\text{m/s}]$